Coding & Solution

Date	4 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Coding & Solution

This document summarizes the implementation of the AI Powered Debt Relief & Financial Recovery Platform. The solution is developed using React.js, FastAPI, SQLite, SQL Alchemy ORM, and Google Gemini API. The project follows modular architecture, clean coding practices, reusable components, and proper documentation.

Instructions

1. Ensure the code is modular, reusable, and follows clean coding standards.
2. Implement authentication, loan management, AI analysis, and dashboard features.
3. Include all required source files, configuration files, and setup instructions.
4. Maintain the source code in a GitHub repository with version control.

Solution Summary

Field	Details
Repository Link / URL	GitHub Repository (Team Project)
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	FastAPI, React.js, SQLAlchemy ORM
Key Features Implemented	User Authentication, Loan Management, Financial Dashboard, AI Settlement Recommendation, Negotiation Letter, History
Pending / Incomplete Features	Gemini API Key configuration and final deployment
Setup / Run Instructions	Install dependencies, run FastAPI backend with Uvicorn, then run React frontend using Vite.

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments /documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The project follows a structured architecture with separate frontend and backend modules. Reusable services, routers, models, and database components improve maintainability and scalability. The application has been tested locally and is ready for further deployment after production API configuration.